

CROP GROWTH SIMULATION COMBINATION OF MILESTONE 1 AND 2

GROUP 2 ABE

```
#include <iostream>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
class CropSystem {
```

```
private:
```

```
    double soilMoisture;
```

```
    double temperature;
```

```
    double sunlight;
```

```
    double cropHeight;
```

```
public:
```

```
    // Constructor (initial values from Milestone 1)
```

```
    CropSystem() {
```

```
        soilMoisture = 50.0;
```

```
        temperature = 25.0;
```

```
        sunlight = 8.0;
```

```
        cropHeight = 5.0;
```

```
    }
```

```
    // Option 1: Automatic update (Milestone 1 style)
```

```
    void autoUpdate() {
```

```
        double rainfall = 5.0;
```

```
        double evaporation = 2.0;
```

```

double usage = 3.0;

soilMoisture = soilMoisture + rainfall - evaporation - usage;

if (soilMoisture < 0) {
    soilMoisture = 0;
}
}

// Option 2: User input update (Milestone 2 style)
void manualUpdate() {
    double rainfall, evaporation, usage;

    cout << "\nEnter rainfall: ";
    cin >> rainfall;
    cout << "Enter evaporation: ";
    cin >> evaporation;
    cout << "Enter water usage: ";
    cin >> usage;

    soilMoisture = soilMoisture + rainfall - evaporation - usage;

    if (soilMoisture < 0) {
        soilMoisture = 0;
    }
}

```

```

// Growth calculation (from Milestone 1)
void growCrop() {
    double r_max = 0.8;
    double M_opt = 60.0;
    double T_opt = 25.0;
    double S_opt = 10.0;

    double f_m = soilMoisture / M_opt;
    double f_t = 1 - ((temperature - T_opt) / T_opt) * ((temperature - T_opt) / T_opt);
    double f_s = sunlight / S_opt;

    if (f_t < 0) f_t = 0;

    double growth = r_max * f_m * f_t * f_s;

    // Extra control logic (Milestone 2)
    if (soilMoisture < 30) {
        cout << "? Low soil moisture! Growth reduced.\n";
        growth *= 0.5;
    }

    cropHeight += growth;
}

// Display results

```

```

void display(int day) {
    cout << fixed << setprecision(2);
    cout << "\nDay " << day << endl;
    cout << "Soil Moisture: " << soilMoisture << "%" << endl;
    cout << "Crop Height: " << cropHeight << " cm" << endl;
    cout << "-----" << endl;
}
};

```

```

int main() {
    CropSystem crop;
    int choice;

    cout << "Crop Growth Simulation System\n";
    cout << "1. Automatic Simulation (Milestone 1)\n";
    cout << "2. Manual Simulation (Milestone 2)\n";
    cout << "Enter choice: ";
    cin >> choice;

    for (int day = 1; day <= 10; day++) {

        if (choice == 1) {
            crop.autoUpdate(); // fixed values
        } else {
            crop.manualUpdate(); // user input
        }
    }
}

```

```
        crop.growCrop();  
        crop.display(day);  
    }  
  
    return 0;  
}v
```